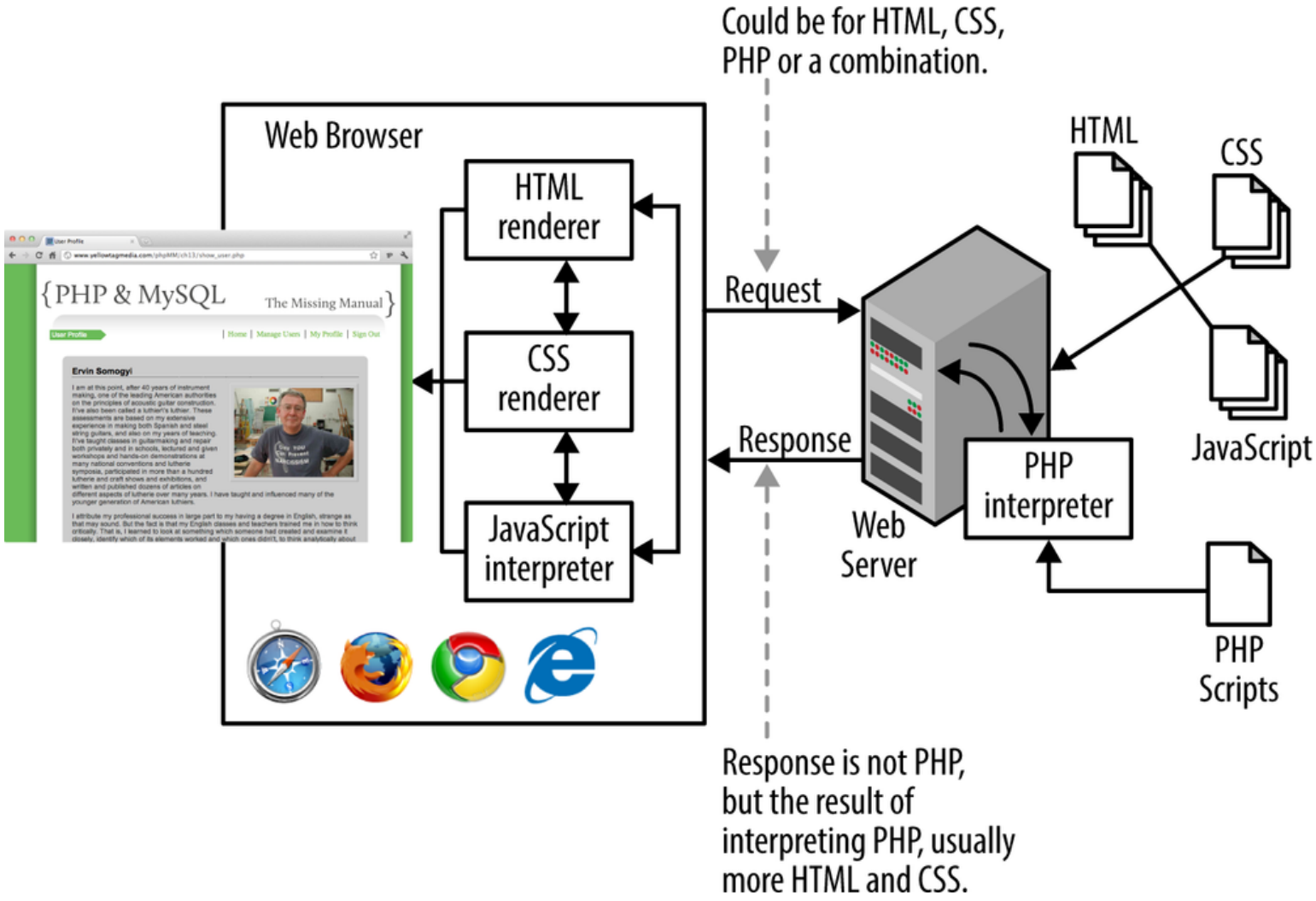
PHP INTRODUCTION 2025

Silvio Priem Mendes, PhD

WAMP/HTTP Architecture



- **DataBase Server not represented at this point**

PHP Introduction

- PHP is a server side scripting language that is embedded in HTML. It is used to manage dynamic content, databases, session tracking, even build entire webapps.
- It is integrated with a number of popular databases, including MySQL, PostgreSQL, Oracle, and Microsoft SQL Server.
- PHP is fast in its execution, especially when compiled as an Apache module.
- PHP supports a large number of major protocols such as POP3, IMAP, and LDAP.
- PHP is forgiving: PHP language tries to be as forgiving as possible.
- PHP Syntax is C-Like.

PHP Common uses

- PHP performs system functions, i.e. from files on a system it can create, open, read, write, and close them.
- PHP can handle forms, i.e. gather data from files, save data to a file, through email you can send data, return data to the user.
- You add, delete, modify elements within your database through PHP.
- Access cookies variables and set cookies.
- Using PHP, you can restrict users to access some pages of your website.
- It can encrypt data.

PHP Scripts

- Typically file ends in **.php** (web server configuration);
- Separated in files with the **<?php … ?>** tag;
- PHP code can make up an entire file, or can be contained in html; this is a choice for now….more on later regarding MVC;
- A PHP block can be placed anywhere in the document;
- Program lines end in **";"** or you get an error;
- Server recognizes embedded script and executes the PHP;
- Result is passed to browser, PHP source isn't visible.

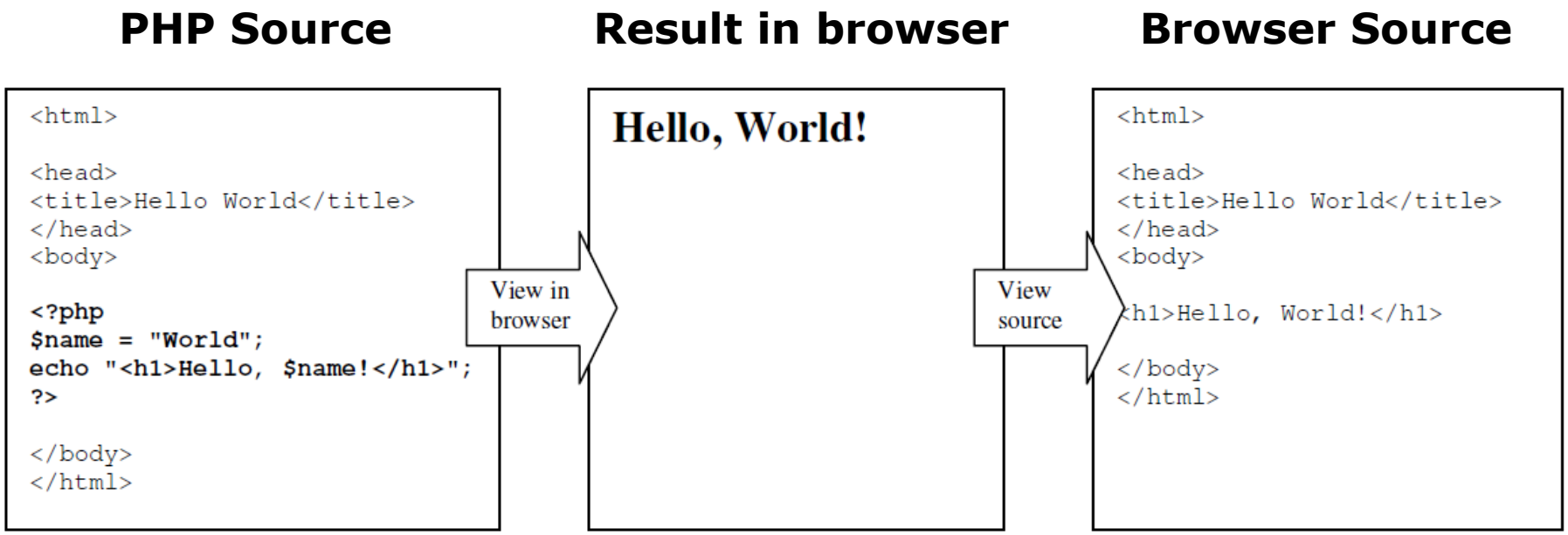
'Hello World' Script in PHP

- Example includes comments

```
<!DOCTYPE html>
<html>
  <body>
    <h1>My first PHP page</h1>
    <?php
          echo 'Hello World!';
          // This is a single-line comment
          # This is also a single-line comment
          /*
              This is a multiple-lines comment
              block that spans over multiple
              lines
          */
     ?>

</body>
</html>
```

"Hello World" Script in PHP



PHP Comments

- A comment in PHP code is a line that is not read/executed as part of the program.
  - Its only purpose is to be read by someone who is looking at the code.

- **Comments can be used to:**
- Let others understand what you are doing
- Remind yourself of what you did - Most programmers have experienced coming back to their own work a year or two later and having to re-figure out what they did. Comments can remind you of what you were thinking when you wrote the code

PHP Case Sensitivity

- In PHP, all keywords (e.g. if, else, while, echo, etc.) and built/in functions, **are NOT case-sensitive**.
- **However; all variable names are case-sensitive.**
  - In the example below, only the first statement will display the value of the $color variable (this is because $color, $COLOR, and $coLOR are treated as three different variables):

```
<!DOCTYPE html>
<html>
<body>
<?php
   $color = 'red';
   echo 'My car is ' . $color . '<br>';
   echo 'My house is ' . $COLOR . '<br>';
   echo 'My boat is ' . $coLOR . '<br>';
?>
</body>
</html>
```

Declaring PHP Variables

• In PHP, a variable starts with the $ sign, followed by the name of the variable:

```
<?php
    $txt = 'Hello world!';
    $x = 5;
    $y = 10.5;
?>
```

- After the execution of the statements above, the variable **$txt** will hold the value **Hello world!**, the variable **$x** will hold the value **5**, and the variable **$y** will hold the value **10.5**;
- **Note:** Unlike other programming languages, PHP has no command for declaring a variable. It is created the moment you first assign a value to it.

PHP Variables

•A variable can have a short name (like x and y) or a more descriptive name (age, carname, total_volume) **RECOMMENDED**.

Rules for PHP variables:

•A variable starts with the **$** sign, followed by the name of the variable

•A variable name must start with a letter or the underscore character

•A variable name cannot start with a number

•A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)

•Variable names are case-sensitive ($age and $AGE are two different variables)

PHP Variables

- **PHP is a Loosely Typed Language**

- In the previous example, notice that we did not have to tell PHP which data type the variable is.

- PHP automatically converts the variable to the correct data type, depending on its value.

PHP Data Types

- Variables can store data of different types, and different data types can do different things.

- **PHP supports the following data types:**
  - String
  - Integer
  - Float (floating point numbers - also called double)
  - Boolean
  - Array
  - Object
  - NULL

PHP String

- A string is a sequence of characters, like "Hello world!".
- A string can be any text inside quotes. You can use single or double quotes (**although there are differences**):

```
<?php
  $x = "Hello world!";
  $y = 'Hello world!';

  echo $x;
  echo "<br>";
  echo $y;
?>
```

PHP String

- Single Quoted

```
$color = 'Green';
```

- Double Quoted **(strings enclosed by " are interpreted)**

```
// outputs Green Eggs and Ham
$food = "$color Eggs and Ham";
echo $food;
```

PHP Integer

- An integer data type is a non-decimal number between -2,147,483,648 and 2,147,483,647.

- **Rules for integers:**
  - An integer must have at least one digit
  - An integer must not have a decimal point
  - An integer can be either positive or negative
  - Integers can be specified in three formats: decimal (10-based), hexadecimal (16-based - prefixed with 0x) or octal (8-based - prefixed with 0)

PHP Integer

- In the following example $x is an integer.
- The PHP **var_dump()** function returns the data type and value:

```
<?php
  $x = 5985;
  var_dump($x);
?>
```

PHP Float

- A float (floating point number) is a number with a decimal point or a number in exponential form.
- In the following example $x is a float.
- The PHP **var_dump()** function returns the data type and value:

```
<?php
  $x = 10.365;
  var_dump($x);
?>
```

PHP Boolean

- A Boolean represents two possible states: TRUE or FALSE.

```
$x = true;
$y = false;
```

- Booleans are often used in conditional testing.
- You will learn more about conditional testing in a later chapter of this course.

PHP Object

- An object is a data type which stores data and information on how to process that data.
  - **In PHP, an object must be explicitly declared.**
  - First we must declare a class of object. For this, we use the class keyword. A class is a structure that can contain properties and methods:

```
<?php
class Car {
    function Car() {
        $this->model = 'VW';
    }
}
// create an object
$herbie = new Car();
// show object properties
echo $herbie->model;
?>
```

- You will learn more about objects in a later chapter of this course.

PHP NULL

- Null is a special data type which can have only one value: **NULL**.
- A variable of data type NULL is a variable that has no value assigned to it.
  - **Note:** If a variable is created without a value, it is automatically assigned a value of NULL.
- Variables can also be emptied by setting the value to NULL:

```
<?php
  $x = 'Hello world!';
  $x = null;
  var_dump($x);
?>
```

PHP Operators

- Operators are used to perform operations on variables and values.

- **PHP divides the operators in the following groups:**
  - Arithmetic operators
  - Assignment operators
  - Comparison operators
  - Increment/Decrement operators
  - Logical operators
  - String operators
  - Array operators (explained later)

PHP Arithmetic Operators

- The PHP arithmetic operators are used with numeric values to perform common arithmetical operations, such as addition, subtraction, multiplication etc.

Operator	Name	Example	Result
+	Addition	$x + $y	Sum of $x and $y
-	Subtraction	$x - $y	Difference of $x and $y
*	Multiplication	$x * $y	Product of $x and $y
/	Division	$x / $y	Quotient of $x and $y
%	Modulus	$x % $y	Remainder of $x divided by $y
**	Exponentiation	$x ** $y	Result of raising $x to the $y'th power (Introduced in PHP 5.6)

PHP Assignment Operators

- The PHP assignment operators are used with numeric values to write a value to a variable.
  - The basic assignment operator in PHP is "=". It means that the left operand gets set to the value of the assignment expression on the right.

Assignment	Same as...	Description
x = y	x = y	The left operand gets set to the value of the expression on the right
x += y	x = x + y	Addition
x -= y	x = x - y	Subtraction
x *= y	x = x * y	Multiplication
x /= y	x = x / y	Division
x %= y	x = x % y	Modulus

PHP Comparison Operators

- The PHP comparison operators are used to compare two values (number or string):

Operator	Name	Example	Result
==	Equal	$x == $y	Returns true if $x is equal to $y
===	Identical	$x === $y	Returns true if $x is equal to $y, and they are of the same type
!=	Not equal	$x != $y	Returns true if $x is not equal to $y
<>	Not equal	$x <> $y	Returns true if $x is not equal to $y
!==	Not identical	$x !== $y	Returns true if $x is not equal to $y, or they are not of the same type
>	Greater than	$x > $y	Returns true if $x is greater than $y
<	Less than	$x < $y	Returns true if $x is less than $y
>=	Greater than or equal to	$x >= $y	Returns true if $x is greater than or equal to $y
<=	Less than or equal to	$x <= $y	Returns true if $x is less than or equal to $y

PHP Inc/dec Operators

- The PHP increment operators are used to increment a variable's value.
- The PHP decrement operators are used to decrement a variable's value.

Operator	Name	Description
++$x	Pre-increment	Increments $x by one, then returns $x
$x++	Post-increment	Returns $x, then increments $x by one
--$x	Pre-decrement	Decrements $x by one, then returns $x
$x--	Post-decrement	Returns $x, then decrements $x by one

PHP Logical Operators

- The PHP logical operators are used to combine conditional statements.

Operator	Name	Example	Result
and	And	$x and $y	True if both $x and $y are true
or	Or	$x or $y	True if either $x or $y is true
xor	Xor	$x xor $y	True if either $x or $y is true, but not both
&&	And	$x && $y	True if both $x and $y are true
||	Or	$x || $y	True if either $x or $y is true
!	Not	!$x	True if $x is not true

PHP String Operators

- PHP has two operators that are specially designed for strings.

Operator	Name	Example	Result
.	Concatenation	$txt1 . $txt2	Concatenation of $txt1 and $txt2
.=	Concatenation assignment	$txt1 .= $txt2	Appends $txt2 to $txt1

PHP Type Coercion

- Just like javascript, php is loosely typed;
- Coercion occurs the same way;
  - If you concatenate a number and string, the number becomes a string;

```
<?php
  $foo = '1';  // $foo is string (ASCII 49)
  $foo *= 2;   // $foo is now an integer (2)
  $foo = $foo * 1.3;  // $foo is now a float (2.6)
  $foo = 5 * '10 Little Piggies'; // $foo is integer (50)
  $foo = 5 * '10 Small Pigs';     // $foo is integer (50)
?>
```

PHP Constants

• Constants are like variables except that once they are defined they cannot be changed or undefined.

- A valid constant name starts with a letter or underscore (no $ sign before the constant name).
- **Note:** Unlike variables, constants are **automatically global** across the entire script.

```
<?php
  define('GREETING', 'Welcome to IPLEIRIA!');
  echo GREETING;
?>
```

BIBLIOGRAPHY